1. Directed Graphs and Their Transposes

Original Directed Graph:
Vertices: {1, 2, 3}
Edges: 1→2, 2→3, 1→3

Transposed Graph: Reverse all edge direction
Edges: 2→1, 3→2, 3→1

2. Undirected Graphs and Their Inverse Graphs Original

~~Undire~~ Undirected Graph:
Vertices: {A, B, C, D}
Edges: {AB, BC, CD}

Inverse Graph: Add edges that don't exist
Original edges: AB - BC, CD
Inverse edges: AC, AD, BD (all pairs not connected).

3. Dense Original → Inverse
If the original is dense (many edges), the inverse becomes sparse (few edges).
Example: Complete graph K_4, ~~the~~ has no inverse graph (all vertices connected mean no missing edges).

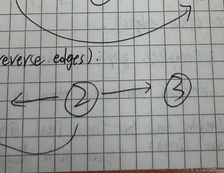
4. Undirected Graphs ~~and~~ ~~Tok~~ Their Dual Graphs
Original Planar Graph: Square with diagonal.

Dual Graph: One vertex per face, edges between ~~outside~~ adjacent faces.

5. Why Dual Only Defined for Planar Graphs?
The dual requires dividing the plane into distinct face, which only works for planar graphs (can be drawn without edge crossings)
Non-planar example: K_5.
① Complete graph on 5 vertices
② Cannot be drawn without edge crossing → no clear face ~~stru~~ structure → no dual possible.

~~Bran~~ Bronkerbosch Execution
like:
1. Directed Graph & Transpose:

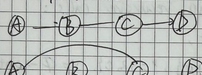
Original:



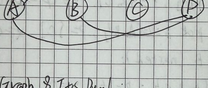
Transpose (reverse edges).

2. Undirected Graph & Inverse

Original:



Inverse:

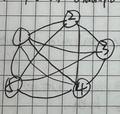


4. Planar Graph & Its Dual

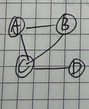
Original:

Its dual:

5. Non-planar example: K_5



Graph:



1. Initial:
$R = \emptyset$, $P = \{A, B, C, D\}$, $X = \emptyset$
Pick A → recurse with $P \cap N(A) = \{B, C\}$.

2. First two recursive calls:
When $v = A$;
$R = \{A\}$, $P = \{B, C\}$, $X = \emptyset$ → continue
The pick B: $R = \{A, B\}$, $P = \{C\}$, $X = \emptyset$
When $v = B$ (nested):
$R = \{A, B, C\}$; $P = \emptyset$, $X = \emptyset$
(Report clique $\{A, B, C\}$)

3. All maximal cliques:
$\{A, B, C\}$ – size 3 ← maximum
$\{C, D\}$ – size 2.

Why:
① $\{A, B, C\}$: all three pairwise connected (edges AB, BC, CA)
② $\{C, D\}$: only two vertices, but no other vertex adjacent to both.
③ $\{A, B, C, D\}$ is not a clique (no edges A-D, B-D)